AEGIS: Adaptive, Evidence-Grounded Interactive Search for Video Moment Retrieval

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Abstract. Timed interactive video retrieval requires fast candidate exploration under penalties for false submissions. We introduce AEGIS (Adaptive Evidence-Grounded Interactive Search), an interactive video retrieval system for the Video Browser Showdown (VBS 2027). AEGIS combines Tencent WeMM-Embedding-4B with Qdrant HNSW indexing and weighted reciprocal rank fusion; it provides entity-aware conversational refinement, parallel VQA with abstention, and an intra-video timeline explorer. Its workflows retain source/frame identity and link answers to the supplied frame evidence; unavailable evidence or invalid output triggers abstention, not a guarantee of factual correctness. Archived outputs from a 1,000-query V3C suite provide a limited retrieval record, not end-to-end VQA measurements or official DRES scores. Their indexed-text overlap and unverified encoder/index identity limit interpretation. The project page (`https://aegis.mncuchiinhuttt.dev/`) provides supplementary specifications and will announce the public code release.

Keywords: Interactive Video Retrieval · Video Browser Showdown · Conversational Search · Evidence-Grounded VQA

1 Introduction and System Contribution

The Video Browser Showdown (VBS) evaluates interactive retrieval under a task clock, with segment or answer submission to the Distributed Retrieval Evaluation Server (DRES) [1,2]. VBS 2027 comprises textual, conversational and visual known-item search (KIS-T, KIS-C, KIS-V), ad-hoc video search (AVS), and visual question answering (VQA) over V3C, Marine Video Kit and laparoscopic gynecology videos [3,4,5,6]. This competition scope does not establish our index coverage. PraK, NII-UIT, Exquisitor and vitrivr combine complementary retrieval and interaction strategies [7,8,9,10]. AEGIS provides the following integrated retrieval and inspection mechanisms:

1. **Source-aware inspection.** Retained candidate payloads link results to indexed frames and source videos. Scene deduplication limits repetitive-frame clutter in candidate lists.
2. **Conversational refinement.** History-aware rewriting, heuristic clarification and explicit relevance feedback connect operator responses to subsequent retrieval.
3. **Evidence-aware VQA and browsing.** Answers carry the frame evidence actually supplied to the model; unavailable evidence and invalid responses are exposed before further inspection or submission.

2 AEGIS System Architecture

Figure 1 shows the AEGIS reference implementation; the public code release will be announced on the project page. Each backend process maintains one operator session; multiple operators require separate instances or session isolation.

Offline records. Ingestion prefers official V3C shots/keyframes, otherwise using local detection and selection, including TransNetV2 [11]. Records retain available source/frame/time identity, captions, narratives, OCR and object labels. Legacy captions can be video-title text rather than generated frame descriptions. ASR is imported or generated with faster-whisper [12], an implementation of Whisper [13]; legacy records need not contain every field. Qdrant stores vectors and searchable `text_blob` payloads in a configurable visual collection [14]. Primary query/index identities (model, revision, dimension and embedding recipe) must match before search. The archived run does not identify its active collection. Audio and shot-parent collections are not default fusion lists.

Models. WeMM-Embedding-4B [15] is the primary encoder. The wrapper uses its native embedding API, L2 normalizes and defaults to 2,560 dimensions; only checkpoint-supported MRL sizes are accepted. Optional SigLIP is disabled by default and needs matching named vectors.

Role	Component or interface
Primary encoder	`tencent/WeMM-Embedding-4B` [15]
SigLIP (optional)	`google/siglip-so400m-patch14-384` [16]
VLM module	Replaceable text/image-to-text adapter
Optional detector	YOLOE [17]; configured filename: `yoloe-26x-seg.pt`
OCR	PaddleOCR [18]; recognition checkpoint not pinned

Table 1. AEGIS components and the configurable VLM interface.

The configurable VLM adapter handles text-only rewriting, expansion and clarification, and frame-image VQA with a JSON response contract. Its backend and answer-generation quality are not established by the archived diagnostics.

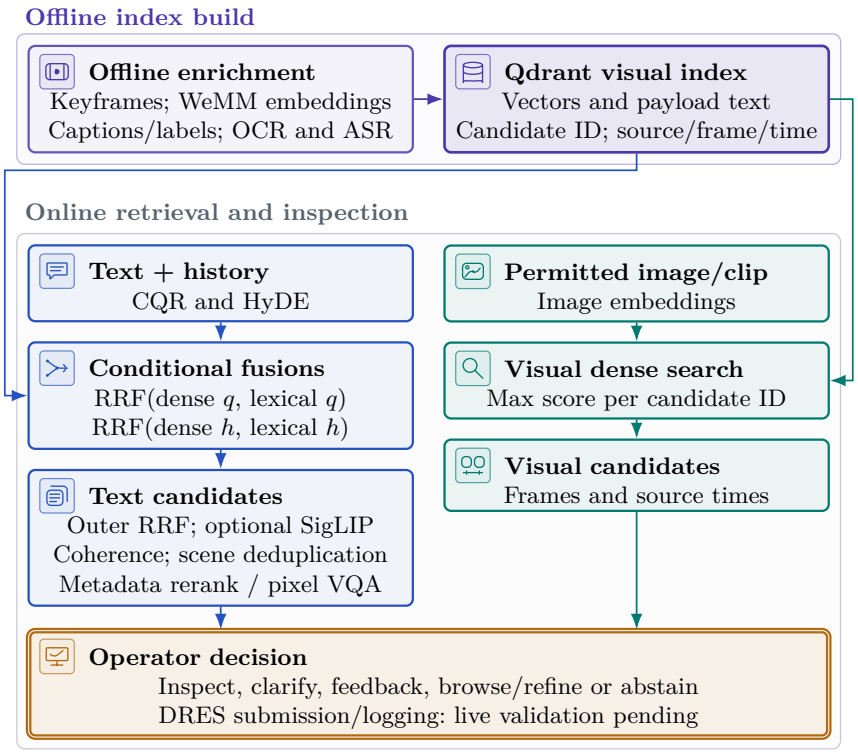


Fig. 1. AEGIS architecture. Text uses weighted, nested RRF; visual queries bypass expansion. Optional paths and live DRES validation remain unmeasured.

2.1 Retrieval, Fusion and Search Effort

Rewritten query q and HyDE expansion h [19] enter separate hybrid calls, each combining primary dense results with an available Qdrant `MatchText/scroll` payload-text list. Scroll order supplies lexical rank, not BM25 scores. The fusion rule is [20]

$$F(d; L_1, \ldots, L_m) = \sum_{j: d \in L_j} \frac{w_j}{k + \mathrm{rank}_{L_j}(d)}. \tag{1}$$

Ranks are one-based; configurable nonnegative weights default to $w_j = 1$, with $k = 60$. The archived run does not record its effective settings. Inner fusions $F(D_q, L_q)$ and $F(D_h, L_h)$ feed an outer RRF, optionally with SigLIP results for q. The API requests 15 candidates per hybrid call. After outer fusion, same-video neighbors within ten native frames receive a coherence boost with weight 0.3. Scores update in place within each video's candidate order. Scene deduplication keeps one hit per (video, scene). The web text path reranks at most ten candidates using metadata, not frame pixels. Dense retrieval uses HNSW [21]; requests can override its search effort or select exact search. UI requests carry fixed ex-

act/verification flags. VQA uses up to eight generation workers for concurrency-capable API clients; local clients remain serial. The archive does not measure this speedup.

3 Task Workflows and Evidence Handling

KIS-T and KIS-C. KIS-T follows the text path. KIS-C rewriting uses history and accepted/rejected descriptions. For the first $n = \min(10, |C|)$ candidates, D is the number of distinct `source_file` values divided by n (zero for an empty pool). With largest scores $s_1 \geq s_2$, define $S = 1 - \text{clip}((s_1 - s_2)/s_1, 0, 1)$; set $S = 0$ for fewer than two scores or $s_1 \leq 0$. At $A = (D + S)/2 \geq 0.7$ by default, a question is generated from at most five different-video summaries (caption text or source ID). The threshold is configurable; these signals are uncalibrated heuristics. The operator manually relays questions to the task giver and enters replies; there is no oracle API. Each clarification turn reruns retrieval before feedback scoring. Its lexical overlap is $o = \min(1, u + 0.25p)$: u is reply-token recall after filtering stopwords/short words; p is the fraction of bigrams/trigrams formed from those filtered tokens and literally found in candidate text (empty overlaps are zero). Reply overlap adds $0.5\, o\, s_{\max}$ only to candidates represented in the question prompt, where $s_{\max}$ is the positive pre-boost maximum, or 1 otherwise. Empty prior IDs leave scores unchanged. Accepted/rejected descriptions enter subsequent rewriting; vector feedback uses normalized Rocchio coefficients $(1, 0.75, 0.15)$.

VQA. A real image or source-video frame is resolved inside the dataset root, preferring frame index over timestamp. The highest-confidence detected box is optionally cropped; failed/absent detection leaves the full frame. This heuristic does not establish relevance to the question. Without media, no VLM answer call is made. The JSON contract contains `found`, `answer`, `confidence` and `reason`. Malformed JSON, `found=false`, empty/sentinel answers, non-finite confidence or values outside $[0, 1]$ cause internal `UNKNOWN`, displayed as `N/A` with a reason. Valid JSON and provenance do not guarantee a correct answer. Confidence is self-reported, with no positive minimum threshold; optional model-based yes/no checks are not an independent oracle. Without those checks, ranking combines 0.4 times retrieval score and 0.6 times VQA confidence. Abstention zeroes the latter, not necessarily the entire score. The answer retains candidate/source identity, resolved frame/time and evidence availability for operator inspection. `UNKNOWN` is an internal state, not a validated DRES response or an enforced submission block.

KIS-V. At most eight uniformly spaced uploaded clip frames are embedded and searched sequentially; repeated candidate IDs retain maximum similarity. An image-query endpoint is also available. Both bypass rewriting/HyDE and require a permissible digital query, not recording the projected presentation [3]. Otherwise, visual matching relies on manual inspection.

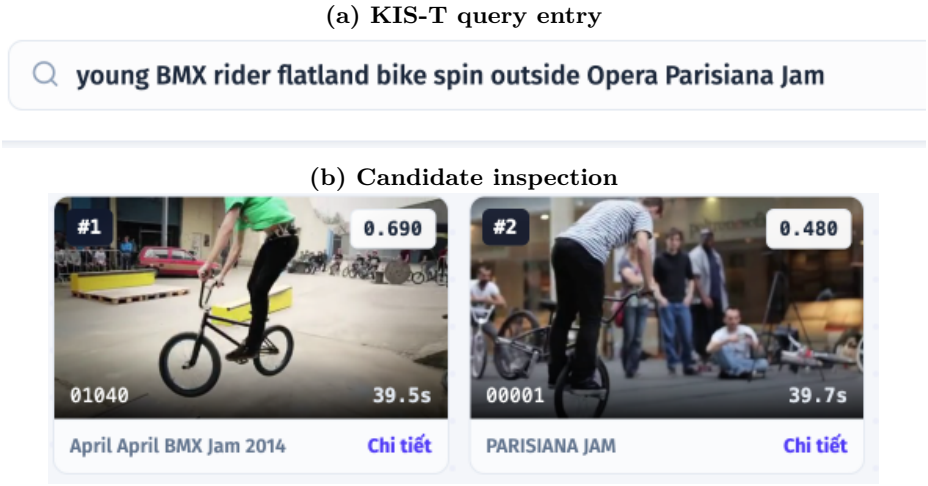


Fig. 2. Archived KIS-T workspace excerpts: (a) query entry; (b) candidate cards with source IDs, timestamps and detail actions. Scores illustrate the interface; diagnostic scope and DRES integration status are described in Sections 4 and 3.1.

AVS and browsing. AVS uses scene deduplication, manual submission and an overridable duplicate-video warning. Timeline browsing defaults to ±30 s around a supplied center for timestamped frames; untimed frames remain and the VQA UI requests 35 s. Browse and in-video search operate on indexed frames, not continuous sub-shot coverage. Figure 2 shows the query controls and source/time candidate cards used for inspection.

3.1 DRES Integration Status

The reference UI sends only a media item and timestamp, including for VQA; answer text and explicit temporal ranges are not wired. AVS submits individual cards. The backend forwards caller-built payloads through generic DRES adapters, but task schemas and live acceptance remain unverified. Some searches, feedback actions and successful submissions are logged. Browsing, playback, early empty-result search returns and failed API submissions are omitted. Records lack team/user identity, operation duration and structured VQA answer or evidence fields. Complete event coverage and remote log acceptance remain integration gaps.

4 Evaluation Scope and Limitations

4.1 Archived Retrieval Diagnostics

An archived development suite contains 1,000 templates (400 KIS-T, 200 KIS-C, 200 VQA, 125 AVS, 75 KIS-V). Its generator extends a 200-query base; added

Hit rule	R@1	R@5	R@10	R@20	MRR
Task-rule	70.4	76.0	77.6	80.1	0.7289
Video-only	70.4	76.3	–	–	0.7299

Query timer	Mean	p_{50}	p_{95}
Recorded milliseconds	476.6	474.6	516.1

Table 2. Archived diagnostics ($N = 1{,}000$). R@K percent, MRR unitless; – means not stored. The timer covers search/local task logic, not VLM, UI or DRES. These are not independently validated effectiveness estimates.

templates and KIS-C replies reuse target captions/`text_blob` already indexed, including video-title text. This creates lexical leakage, not independent evidence of visual discrimination. Table 2 transcribes the retained saved aggregate, not a full-system run.

The historical runner assigns `final_rank` via normalized substring ID matching. R@K is the fraction with rank $\leq K$; MRR averages reciprocal rank (miss=0). KIS-C/KIS-V use video hits; AVS accepts any listed target video; VQA uses an exact target-point ID or first target-video match, not answer correctness. KIS-T uses video rank one, otherwise temporal rank, then video rank. Temporal tests accept differences ≤ 600 native frames when both indices exist, otherwise ≤ 24 s when both timestamps exist; without either complete pair, video matching alone qualifies. These separate script criteria do not constitute pooled temporal retrieval accuracy.

KIS-V replay uses text, not reference images; KIS-C uses prefilled query/reply fields rather than live dialogue. No VLM is called. VQA flags use text/video-rank proxies and automatic negative-case passes, not generated-answer or refusal assessment. The aggregate lacks per-query ranks, independent labels and encoder/checkpoint/index identities; compatibility and end-to-end effectiveness remain unverified.

5 Conclusion

AEGIS integrates WeMM-based multimodal retrieval, conversational refinement, relevance feedback, frame-linked answer inspection and intra-video browsing in a common operator workflow. Archived outputs are limited to the retrieval path and local task logic described in Section 4. Further validation requires matched model/index versions, independent temporal labels and repeated replay; DRES rehearsal and collection-level validation remain deployment requirements.

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